UNIT- 1

FUELS

Abstract

Energy is necessary to do work. Energy is produced when fuel undergoes combustion. This unit deals with different types of calorific values, knocking in petrol and diesel engine, knocking in terms of octane no and cetane no and solutions to overcome adverse effect of knocking. As it is known after few years coal and petroleum will not be available, so we will be discussing some inexhaustible source of energy. A special category of fuel and oxidizer is required to undergo combustion to give high thrust to rocket. Briefly we will discuss about the rocket propellant types, characteristics and properties.

Blow-up Syllabus

Sl No	Topic to be taught	Duration
1	Introduction to Thermochemistry, calorific value of fuels	1 hour
2	Numericals on calorific values, knocking in internal combustion engines	1 hour
3	Reasons for knocking in IC engine, Knocking in disel engine	1 hour
4	Octane and Cetane number	1 hour
5	Antiknocking agents. Biodiesel-synthesis, power alcohol	1 hour
6	Green fuel- hydrogen production and storage	1 hour
7	Rockets Fuels: Properties	1 hour
8	Characteristics of Rocket Propellant and types	1 hour

UNIT- I

Thermochemistry

Thermochemistry deals with the study of heats of chemical reactions. When a chemical reaction takes place, energy in various forms may either be emitted or absorbed. During many reactions the temperature rises indicating that heat is being evolved. In others, when temperature falls, absorption of heat is indicated. Thus, the study of the heat produced or required by chemical reaction forms the basis of thermochemistry. The reactions in which heat escapes from the system to the surroundings are termed as exothermic process that means the surrounding ultimately becomes hotter. The release of heat leads to decrease in the enthalpy of a system when the pressure remains constant, thus the enthalpy change is negative. The chemical processes in which heat is absorbed by the system from the surroundings are endothermic process that means the surrounding ultimately become colder. Here the absorption of heat leads to increase in the enthalpy of a system at constant pressure, thus the enthalpy change is positive. A fuel is defined as naturally occurring or artificially manufactured combustible carbonaceous material which serves particularly as source of heat and light and also in few cases as a source of raw material.

Fuel: A fuel is the substance which on combustion produces a large amount of heat.

$$\text{Fuel} + O_2 \xrightarrow{\textit{high temp}} \text{Combustion products} + \text{Heat}$$

As most of the fuels contain carbon or carbon and hydrogen, the combustion involves the oxidation of carbon-to-carbon dioxide and hydrogen to water. Sulphur, if present, is oxidized to sulphur dioxide while the mineral matter forms the ash.

Importance of hydrocarbons as fuels:

Fossil fuels contain mainly hydrocarbons. These hydrocarbons are important sources of energy in daily life. Hydrocarbons are used as energy sources in cooking, lighting, automobiles, production of electricity in thermal power plants etc. These hydrocarbon fuels meet 80% of the world's energy demand. Thus, hydrocarbons are important sources of energy.

Characteristics of a Good/ideal fuel:

While selecting an ideal fuel for domestic or industrial purpose we should keep in mind that

the fuel selected must possess the following characteristic properties.

1. It should possess high calorific value.
2. It should have proper ignition temperature. The ignition temperature of the fuel should neither be too low nor too high.
3. It should not produce poisonous products during combustion. In other words, it should not cause pollution on combustion.
4. It should have moderate rate of combustion.
5. Combustion should be easily controllable i.e., combustion of fuel should be easy to start or stop as and when required.
6. It should not leave behind much ash on combustion.
7. It should be easily available in plenty.
8. It should have low moisture content.
9. It should be cheap.

Classification of fuels: Fuels are classified into two types.

Based on their origin they are classified into

- Primary fuels
- Secondary fuels

Primary Fuels: There are naturally occurring fuels which serves as source of energy without any chemical processing.

Ex: Wood, Coal, Crude oil, Natural gas, Peat, Lignite, Anthracite.

Secondary Fuels: -They are derived from primary fuels & serves as source of energy only after subjecting to chemical processing.

Ex: Charcoal, Coke, producer gas, Petrol, Diesel etc.,

Based on their physical state fuel are classified into

- Solid
- Liquid
- Gaseous fuels

Fuels						
Primary				**Secondary**		
Solid	**Liquid**	**Gas**		**Solid**	**Liquid**	**Gas**
Coal, Wood	Crude Oil (petroleum)	Natural gas		Charcoal, coke	Petrol, Diesel	Coal gas, water gas

Table 1.1: Fuel classification

Calorific value

Calorific value is defined as the amount of heat liberated when a unit mass of fuel is burnt completely in presence of air or oxygen.

Calorific value is of two types as follows:

Higher calorific value. (HCV) or Gross calorific value. (GCV)

Lower calorific value. (LCV) or Net calorific value. (NCV)

HCV: - It is the amount of heat liberated when a unit mass of fuels burnt completely in the presence of air or oxygen and the products of combustion are cooled to room temperature. Here it includes the heat liberated during combustion and the latent heat of steam. Hence its value is always higher than lower calorific value.

LCV: - It is amount of heat liberated when a unit mass of fuel is burnt completely in the presence of air or oxygen and the product of combustion are let off completely into air. It does not include the latent heat of steam. Therefore, it is always lesser than HCV.

Calculation of Heat associated with steam

Let the percentage of hydrogen in the fuel be "H"

2 kg of hydrogen produces 18 kg of water

Therefore, H kg of Hydrogen produces $(18/2)$ X H kg of water

Water produced by 1 kg of fuel = (18/2) x (H/100) kg = 0.09 H kg

Heat associated with steam = weight of water produced x Latent heat of condensation or vapourization

= 0.09H x 587 x 4.2 kJ/kg

(18 is the molecular weight of H_2O and 587 is the latent heat of condensation in calorie per gram)

∴ Gross calorific value = Net Calorific value + Latent Heat of condensation or vapourisation of water produced.

Numerical

1. Find out the percentage of Hydrogen in the 5g of Hydrocarbon fuel burnt in air and heat carried away by H2O is 250kJ/kg

Ans: Heat associated with steam = 0.09H x 587 x 4.2 kJ/kg
H = (Heat associated with steam)/ (587 x 4.2 x 0.09) = 1.126

2. 3.1kg o fuel was burnt in air and combustion products were allowed to cool at 298K. The calorific value obtained by the process is 500 kJ/kg. If the % of hydrogen in the fuel is 2 then what is the net calorific value.

Ans: % of hydrogen = 2
100 kg fuel contains 2 kg of hydrogen
GCV = 500 kJ/kg
NCV = (GCV – (0.09H x 587 x 4.2)) kJ/kg
= (500 – (0.09*2*587*4.2)) kJ/kg
= 56.22 kJ/kg

Knocking of IC Engine:

Knocking in Petrol Engines:

In petrol engines, the mixture of petrol and air is drawn in to the cylinder. The fuel- air mixture is compressed by the piston and is ignited by an electric spark. As the flame front travels in the combustion chamber, rapidly expanding combustion products compress the remaining unburnt fuel and raise its temperature. If the flame front travels rapidly at an optimum speed, the combustion of unburnt fuel takes place smoothly. On the other hand, if the flame front travels too slowly, the entire last portion of fuel mixture may get heated up beyond its ignition temperature and undergo instantaneous explosive combustion. This result in emission of a characteristic rattling sound called “knocking”.

Knocking is the rattling noise or metallic sound produced in an internal combustion engine due to uneven combustion (detonation) of fuel and air mixture.

The reasons for Knocking are

(1) Very high compression ratio causes preignition.

Compression ratio is the ratio of the volume of the fuel and air mixture (V1) at the end of suction stroke to the volume of the mixture (V2) at the end of compression stroke.

Compression ratio (CR) = V1/V2

The value (CR) will always be greater one. Power output of IC engine increases continuously with increase in compression ratio. The compression ration corresponding to maximum power output is known as highest useful compression ratio (HUCR).

Very high compression ratio means that the fuel and air are compressed to a maximum extent, leads to increase in the temperature of the cylinder. Hot spots are formed inside. When the fuel and air mixture are compressed, hot spots ignite the fuel even before spark plug produces spark. This phenomenon is called preignition which leads to knocking.

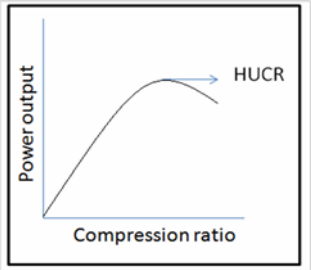


Figure 1.1: Highest useful compression ratio

(2) Peroxide formation leads to self-Ignition

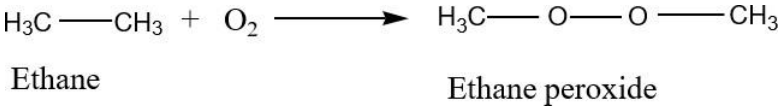
A. **Under Normal** Conditions there is a slow oxidation of the fuel.

Normal combustion takes place due to the chain reaction. The overall reaction may be represented as

$$C_2H_6 + 3/2O_2 \longrightarrow 2CO_2 + 3H_2O$$

B. **Under Knocking Conditions**, the rate of combustion is very high. It involves following steps.

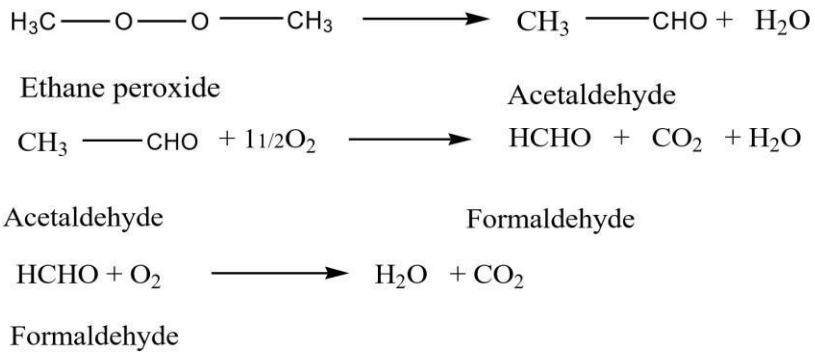
1. Oxygen combines with hydrocarbon molecule forming peroxides.



Ethane — Ethane peroxide

2. The peroxides decompose readily to give a number of gaseous

products. For example,



Fast reaction leads to rapid increase of pressure. This **results in knocking.**

(3) Chemical Structure of the Fuel:

Knocking tendency decreases with the increase in compactness of the molecule and increases with the increase in the length of hydrocarbon chain.

Ex., i) 2, 2-dimethyl pentane knocks lesser than 2-methyl hexane

ii) n-butane knocks lesser than n-hexane.

The tendency to knock has been found to decrease in the order

n-alkane > mono substituted alkanes > cycloalkane > alkenes > Polysubstituted alkanes >aromatic compounds.

Octane Number:

Graham Edger proposed an arbitrary scale for expressing the knocking characteristics of gasoline in combustion engine as octane number in 1972 on the basis of antiknock properties. It has been found that n-heptane, CH_3-CH_2-CH_2-CH_2-CH_2-CH_2- CH_3, knocks very badly and hence, its antiknock value has been arbitrarily given zero. On the other hand, isooctane (2, 2, 4 – trimethylpentane) gives very little knocking, so its antiknock value is given as 100.

Thus, Octane number (or rating) of a gasoline (or any other internal combustion engine fuel) is the percentage of isooctane in a mixture of isooctane and n-heptane, which matches the fuel under test in knocking characteristics.

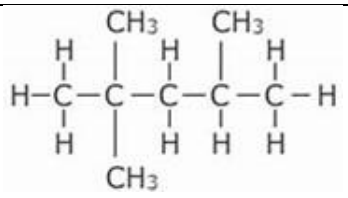
Reference Fuels	Chemical Structure	Octane
Isooctane (2,2,4 –trimethylpentane)		100
n-heptane	CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_3	0

Table 1.2: Reference structures for Octane no scale

Ex: If a sample of petrol gives as much knocking as the mixture of 65% isooctane and 35% n-heptane, then the octane number of petrol is 65.

Anti-Knocking Agents

The octane rating of gasoline samples can be increased by the addition of certain organometallic compounds called anti-knocking agents and the process is called “doping”.

- An extensively used anti-knocking agent is tetraethyl lead Pb(C2H5)4 0.5ml/litre. (but it causes air pollution by Pb.)
- Addition of alkylate fraction to petrol i.e. branched chain alkane of higher mol. wt.
- Addition of benzene, toluene, xylene (BTX) to petrol.
- Addition of methyl t-butyl ether (MTBE) or ethyl t-butyl ether.
- Adjusting suitable compression ratio

Quality of Petrol can be Improved by Adding:

- Antioxidant to improve oxidation resistance.
- Octane no. improvers like MTBE.
- Freezing point depressant like ethanol, glycol to minimize ice formation in fuel line in cold regions.

- Detergents like fatty amines to clean fuel line and carburettor.

Knocking in Diesel Engines

It works on the principle of compression ignition. Air is drawn into the cylinder and compressed to a pressure of 3.5 X 10^{-3} kgm^{-2}. During compression cylinder gets heated around 350 °C. Then diesel is introduced in the form of spray which is spontaneously get ignited and burns

producing pressure on the piston. It takes some time for the fuel to get ignited. The time interval between the fuel injection and ignition is called **ignition delay**. It will be milliseconds. The **shorter ignition delay** leads to burning of the fuel at the rate which it is injected. Whereas **longer ignition delay** results in accumulation of the fuel in the engine and this causes an explosive combustion when ignited. As a result, the **diesel engine knocks**.

Cetane Number of Diesel

Cetane Number is a measurement of the combustion quality of diesel fuel during compression ignition. It is a significant expression of the quality of a diesel fuel. Cetane number is a measure of a fuel's ignition delay, the time period between the start of injection and the first identifiable pressure increases during combustion of the fuel. In a particular diesel engine, higher cetane fuels will have shorter ignition delay periods than lower cetane fuels.

The suitability of a diesel fuel is determined by its cetane value which is the percentage of n-hexadecane in a mixture of n- hexadecane and 2-methyl naphthalene, which has the same ignition characteristics as the diesel sample.

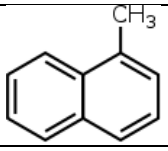
Reference Fuels	Chemical Structure	Cetane Number
α-methyl Naphthelene		0
Cetane (Hexadecane)	CH_3-$(CH_2)_{14}$-CH_3	100

Table 1.3: Reference structures for Cetane no scale

Relative order of cetane no. values

n-alkanes > cycloalkanes > alkenes > branched alkanes > Aromatics

How to increase Cetane No.

Alkyl nitrates (principally 2-ethylhexyl nitrate) and di-tert-butyl peroxide are used as additives to raise the cetane no.

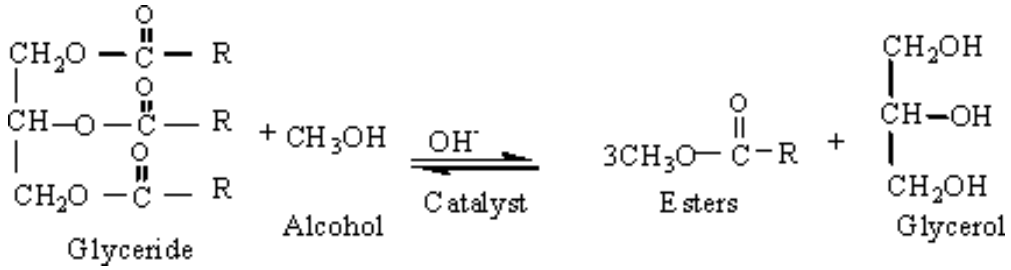
- Antioxidant to improve oxidation resistance during storage.
- Lubricity additives for lubrication of fuel injection system.

- Detergents to clean fuel line and carburettor.

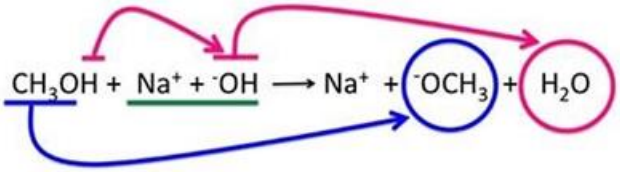
Biodiesel

Chemically biodiesel is the mixture of methyl esters of long chain carboxylic acids. It is eco-friendly fuel obtained from renewable sources like vegetable oils and animal fats. Common vegetable (edible and Non-edible) oils used are Soyabean oil, palm oil, peanut oil, cotton seed oil, corn oil, pongamial oil, Jatropha oil etc. Vegetable oils are the triglycerides, which have high viscosity, flash point and low heating value. Hence they cannot be used as such in diesel engine. Triglycerides are need to be converted into biodiesel by a process is called Transesterification. During the transesterification, the triglyceride is treated with methanol or ethanol in the presence of a base like KOH/NaOH, forms methyl or ethyl esters of fatty acids **(biodiesel)**

Conditions: Temperature: 55-60 C, Catalyst-KOH/NaOH, Reaction time-1-8 hrs.



Separation:



As above two products are immiscible phases, we can easily drain off the glycerine/glycerol by washing with water and biodiesel can be collected.

Extra information:

Compounds present in biodiesel are like,

Methyl palmitate	$H_3C - (CH_2)_{14} - COOCH_3$
Methyl stearate	$H_3C - (CH_2)_{16} - COOCH_3$
Methyl oleate	$H_3C - (CH_2)_7 - CH = CH - (CH_2)_7 - COOCH_3$
Methyl linoleate	$H_3C - (CH_2)_5 - (CH = CH)_2 - (CH_2)_7 - COOCH_3$

Advantages of Biodiesel:

- Biodiesel is cheaper, as it is manufactured from cheap, non-edible or waste oil or animal fats.
- It has high cetane numbers 46 to 54 and high C.V of about 40 kJ/gm.
- It is regenerative and environment friendly.
- It does not give out particulate and CO pollutants, as Oxygen atoms in biodiesel help for complete combustion.
- It has certain extent of lubricity, due to higher oiliness of the esters.
- Its use provides good market to vegetable oils and reduces our dependence on diesel on foreign countries, saving currency.
- It is clean to use biodiesel in diesel engines.

Limitations of Biodiesel:

- Cloud and pour points of biodiesel are higher than diesel and can cause problem in fuel flow line. So it cannot be used in cold regions.
- Biodiesel may have dissolving action rubber hoses, gaskets.
- There is shortage of vegetable oils and the starting material if costly, the biodiesel will be costly.
- Biodiesel strongly adheres on metals and can become gummy.

Power Alcohol

When ethyl alcohol is used as fuel in internal combustion engine, it is called as power alcohol. It contains 25% of alcohol and 75% petrol.

Advantages of Power Alcohol:

- Ethyl alcohol has good antiknocking property and its octane number is 90, while the octane number of petrol is about 65. Therefore, addition of ethyl alcohol to petrol increases its octane number.
- Alcohol has property of absorbing any traces of water if present in petrol.
- If a specially designed engine with higher compression ratio is used, then the disadvantage of lower C.V. of ethyl alcohol can be overcome.

- Ethyl alcohol contains 'O' atoms, which help for complete combustion of power alcohol and the polluting emissions of CO, hydrocarbon, particulates are reduced largely.
- Use of ethyl alcohol in petrol reduces our dependence on foreign countries for petrol and saves foreign currency considerably.
- Power alcohol is cheaper than petrol.

Disadvantages of Power Alcohol:

- Ethyl alcohol has C.V. 7000 cal/gm much lower than C.V. of petrol 11500 cal/gm. Use of power alcohol reduces power output upto 35%.
- Ethyl alcohol has high surface tension and its atomisation, especially at lower temperatures, is difficult causing starting trouble.
- Ethyl alcohol may undergo oxidation to form acetic acid, which corrodes engine parts.
- Ethyl alcohol obtained by fermentation process directly cannot be mixed with petrol but it has to be dehydrated first.
- As ethyl alcohol contains 'O' atoms, the amount of air required for complete combustion of power alcohol is lesser and therefore carburettor and engine need to be adjusted or modified, when only ethyl alcohol is used as fuel.

Green Fuels

Demand for green fuel technologies is rising given their potential to accelerate the transition to more sustainable forms of energy while still supporting current energy models with all their regional variations. Amongst all green fuels hydrogen is considered to be the best alternative. Hydrogen is a zero-emissions source of fuel for trains, buses and cars. It can be used as a feedstock gas for industries such as chemical, refining and steel. In addition, it is a source of heat and power for buildings, and can buffer energy generated from renewable sources.

Hydrogen Production

Over the last decades, an increasing interest has been devoted to the generation of hydrogen from solar-powered water splitting devices as a direct approach to generate a clean and sustainable energy vector. Among the different alternatives, a direct photocatalytic process is the most straightforward, where a light absorber is able to produce hydrogen and oxygen under light irradiation without an external bias.

Photocatalytic water splitting:

The photocatalytic generation of solar fuels by suitable photocatalytic materials suspended in water is certainly one of the most attractive and challenging pathways for decarbonizing current energy systems. From the outside, it seems as a simple approach combining water, solar light, and a semiconductor. However, each of these components plays a key role in the overall efficiency of the reaction, and mostly, many factors including thermodynamic, kinetic, and other requirements hold back this technology from its maximum potential. Photocatalysts must be semiconductors. The minimum requirement is to generate electrons-hole pairs due to irradiation, normally by a UV-vis source. But it must also have a suitable crystalline phase, which ensures the formation of the electron-hole pairs and does not favour its recombination.

Material requirement of photo catalysts

- Band gap: band gap of the materials used in catalysis should be more than 1.23 ev and sufficiently small to make efficient use of solar spectrum (<3eV). Ans it should be having band levels suitable for water splitting.
- High crystallinity: Defects can act as recombination sites
- Long term stability: Charge transfer used for water splitting and not for corrosion

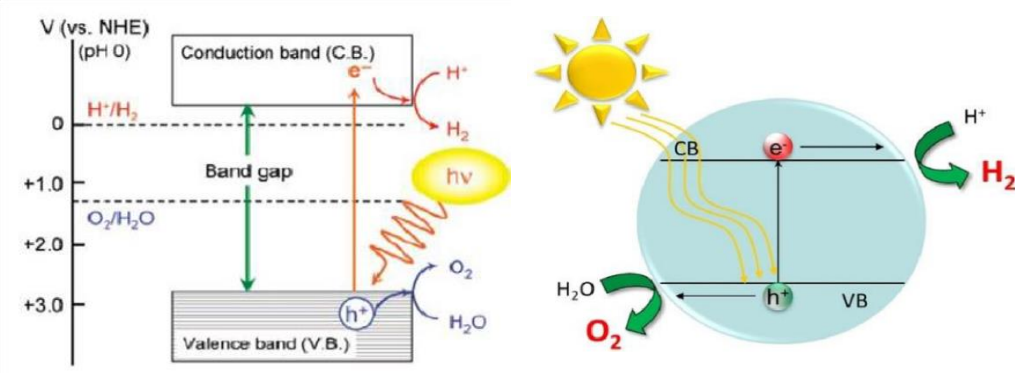


Figure 1.2: Band diagram of catalyst and Hydrogen production by water splitting

Water splitting electrode should be

- Efficient light absorber
- Charge separator
- Charge conductor

Photocatalytic water splitting is an artificial photosynthesis process with photocatalysis in a photoelectrochemical cell used for the dissociation of water into its constituent parts, hydrogen and oxygen, using light.Water splitting electrodes work much like the leaves of the plants during photosynthesis. They absorb energy from the sunlight and use it to generate charge carriers. Negatively charged electrons and positively charged holes. These charges are delivered to water and water molecules swipe electrons away to form hydrogen gas and whereas holes strip the molecules of electrons to form oxygen gas.

Hydrogen Storage

Hydrogen energy storage is an area of growing interest due to the greater storage capacity offered by hydrogen over batteries, as well as the benefits offered by the renewable generation afforded by hydrogen as a fuel. Hydrogen can be stored either as a gas or as a liquid. Hydrogen gas storage typically requires the use of high-pressure tanks (350–700 bar or 5000–10,000 psi), while liquid hydrogen storage requires cryogenic temperatures to prevent it boiling back into a gas (which occurs at −252.8°C). Hydrogen can also be stored on the surface of (adsorption) or within solid materials (absorption).

General method of hydrogen storage

In modern practice hydrogen can be stored in three different ways

1. As a gas under high pressures (in pressure cylinder)
2. In liquid form under cryogenic temperatures (at low temperature storage)
3. On the surface of or within solid and liquid materials (Metal hydrides as well as carbon nanotubes and metal organic frameworks)

Rocket Fuels

Introduction

When you relax in the moonlight away from the din and bustle of a hectic life in a remote village during your vacation, you may look at the sky and feel like going to the moon in your imagination. You may wonder how to travel to outer space. It is not that you are attracted by the beauty of the moon or stars. Rather the charisma of celestial objects like the sun, stars, planets, and moon has impressed several poets and writers to create volumes of literary work since ages. Humans from time immemorial have been baffled by the exotic beauty of the celestial bodies in space. The desire of modern humans to know more about these bodies in space has led to the invention of the telescope, which has been instrumental in unravelling a plethora of knowledge about space. Today, humans are able to send spacecraft even to distant celestial bodies like the moon and mars. If one wants to travel above 25 km into space, one cannot use an air-breathing turbojet because of the nonavailability of sufficient oxygen. One must con- sider using a non-air-breathing propulsive device, which can carry both fuel and an oxidizer. Such a device is known as a rocket engine.

Basic Principle of Propulsion

Recall that propulsion is a method by which an object is propelled in a particular direction. The word “propulsion” stems from the Latin word propellere, where pro means forward or backward and pellere means drive or push. In addition, we know that the verb “propel” means to drive or cause to move an object in a specified direction. Hence, for the study of propulsion we will have to concern ourselves with this propelling force, the motion thereby caused, and the bodies involved. The study of propul- sion is not only concerned with rocket engines but also with vehicles such as aircraft, automobiles, trains, and ships. We may recall that the principle of Newton’s laws of motion is the basis for the theory of jet propulsion. Jet propulsion can be expounded mainly by the second and third laws of motion. For example, a spacecraft is flying vertically at uniform speed. The resultant force in the vertical direction must be zero to satisfy Newton’s second law of motion, according to which an unbalanced force acting on the body tends to produce an acceleration in the direction of the force which is proportional to the product of mass and acceleration. In other words, the spacecraft must produce thrust which must be equal to the drag force caused due to the fluid motion over the body of this spacecraft and the gravitational force. For accelerating the spacecraft, one needs to supply higher thrust

than that of drag forces and gravitational force acting on it. According to Newton's third law of motion, we know that for every acting force, there is an equal and opposite reacting force. The acting force is the force exerted by one body on another, while the reacting force is exerted by the second body on the first. Although these forces have equal magnitude and occur in opposite directions, they never cancel each other because these forces always act on two different objects.

Rocket Propellant

A propellant consists of all the chemical materials, including fuel and oxidizer, along with certain additives necessary for sustaining the combustion process to produce high-pressure hot gases, that which are expanded in a nozzle to produce thrust. Note that the principal ingredients of a propellant are the fuel and the oxidizer. Generally, a fuel is a chemical substance that reacts with an oxidizer while releasing thermal energy. Chemically, fuel can be defined as the chemical substance that can donate electrons during chemical reaction with oxidizer. In contrast, an oxidizer is that chemical substance that accepts electrons during chemical reaction with the fuel. The property of an atom by virtue of which it can either accept or donate electrons is known as electronegativity, which distinguishes a fuel and an oxidizer. Additives are used to enhance the properties of the propellant. We will see more about additives while discussing solid and liquid propellants in subsequent sections. Several kinds of solid, liquid, and hybrid chemical propellants have been devised over the years.

Classification of chemical propellants

Chemical propellants can be classified according to their respective physical states, namely, liquid, solid, gel, or hybrid, as shown in Figure 1. Generally, in solid propellants, fuel and oxidizer are stored and mixed together in a solid form. Note that a solid propellant needs to have a minimum ignition temperature for ignition to occur. Solid propellants can be broadly divided into two categories, namely, homogeneous and heterogeneous, based on the distribution of fuel and oxidizer in the propellant, as shown in Figure 1. In the case of homogeneous propellants, fuel and oxidizer are contained in the same molecule of the propellant. In the case of heterogeneous propellants, solid fuel and oxidizer retain their respective physical identities. Homogeneous propellants are further classified into two categories: (1) single base and (2) double base. Generally, a single- base propellant is a single

compound that contains both fuel and oxidizer. An example of a single-base propellant is nitrocellulose, which has both oxidative and reduction capabilities, whereas, double-base propellants consist of two components. The most widely used double-base propellants are made of nitrocellulose (NC) and nitroglycerine (NG), of course, with a few additives. Heterogeneous propellants are also known as composite propellants, which consist of a fuel and oxidizer mixture. Generally, mineral salt is used as an oxidizer, while polymer and metal powders are used as fuel. For example, ammonium percolate (NH_4ClO_4) as an oxidizer and polybutadienes with metal powders as fuel are commonly used as composite propellant in solid-propellant rocket engines. Liquid propellants have several desirable characteristics over solid propellants and hence are used particularly when higher specific impulse is required.

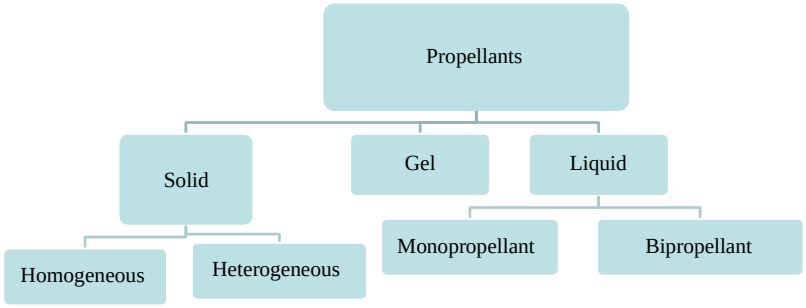


Figure 1.3: Classification of chemical propellants

Based on the mode of ignition, liquid propellants can be classified into two categories:
(1) hypergolic and (2) nonhypergolic. In hypergolic propellants, liquid fuel and oxidizer react spontaneously without external ignition energy. For example, hydrazine can be considered as a hypergolic propellant as it undergoes exothermic reactions and gets decomposed into water and oxygen spontaneously while coming in contact with suitable catalyst even in ambient temperature. But in the case of nonhypergolic propellants, suitable amount of ignition energy is provided to ignite the liquid fuel and oxidizer for combustion to take place. Both hypergolic and nonhypergolic propellants can be divided further into two categories, namely, monopropellants and bipropellants, as shown in Figure 1.3. Under the nonhypergolic propellant category, monopropellants can be further classified into two categories: (1) homogeneous and (2) composite, as shown in Figure 1.3. In the case of nonhypergolic monopropellants, fuel and oxidizer are contained in the same molecule. Methyl nitrate (CH_3NO_3) is an example of a simple nonhypergolic monopropellant. But in the case of composite nonhypergolic

monopropellants, a mixture of fuel and oxidizer is used. Amyl acetate ($C_7H_{14}O_2$) and nitric acid (HNO_3) can be considered as composite nonhypergolic monopropellant. In order to enhance specific impulse with higher metal loading and overcome the problems of sloshing during flight of the liquid-propellant rocket engine, liquid fuel is converted into gel. Generally, gel propellant acts like solid and becomes liquid when subjected to a higher shear rate. A combination of solid and liquid is termed as hybrid propellant.

General characteristics of propellants

All types of chemical propellants based on physical entity have their own desirable characteristics for high-level performance. Each propellant must have higher energetic properties that ensure higher heat release rate, accompanied by high combustion temperature, characteristic velocity, and specific impulse. Besides this, the propellant must have high

ballistic properties, namely, low density, higher ignitability, performance reproducibility, and least combustion instability. Using low-density propellants means that larger storage tanks will be required, thus increasing the mass of the launch vehicle. Other properties, namely, storage stability, less prone to explosion hazard, smokeless exhaust, low price, and easy processing are some of the desirable characteristics of propellants. The common desirable properties of chemical propellants are as follows:

1. Propellant must have high chemical energy release so that it can have higher combustion temperature leading to high characteristic velocity.
2. It can have low molecular weight of combustion product leading to high exhaust velocity Ve and thus can have high specific impulse.
3. It can have a high density such that large amount of chemical energy can be stored in the smallest volume and thus can have a compact design.
4. Easy to ignite even under low-pressure condition.
5. Physically and chemically stable with respect to time.
6. Smoke-free and nontoxic in nature.
7. Easy and expensive to manufacture and handle during operation.
8. Easily available and low price.
9. Less prone to explosion hazard.
10. Low emission level.

Chemical Rocket Engines

In case of chemical rocket engines, chemical energy released during the burning of fuel and oxidizer is used to raise the temperature and pressure of the gas which is expanded in a CD nozzle to produce thrust. Generally, the hot gases at high pressure are accelerated to high supersonic velocities in the range of 1500–4000 m/s for producing thrust. It may be noted that both fuel and oxidizer are being carried along with the engine unlike in air-breathing engines. Based on the physical state of the propellant (fuel and oxidizer), as discussed earlier, chemical rocket engines can be broadly divided into three categories: (1) solid propellant, (2) liquid propellant, and (3) hybrid propellant.

The igniter initiates the combustion process on the surface of the propellant when actuated with the help of an electrical switch. As a result, the propellant grains will start burning and filling the empty combustion chamber, hence building up the chamber pressure. Subsequently, the high temperature and high-pressure gases are expanded in the super- sonic nozzle to produce the requisite thrust. Generally, these nozzles are made of high temperature materials, namely, metals with graphite coating, and are ablative materials that can take a high thermal load with minimal corrosion. The three types of rocket engines design is given below in Figure1.4.

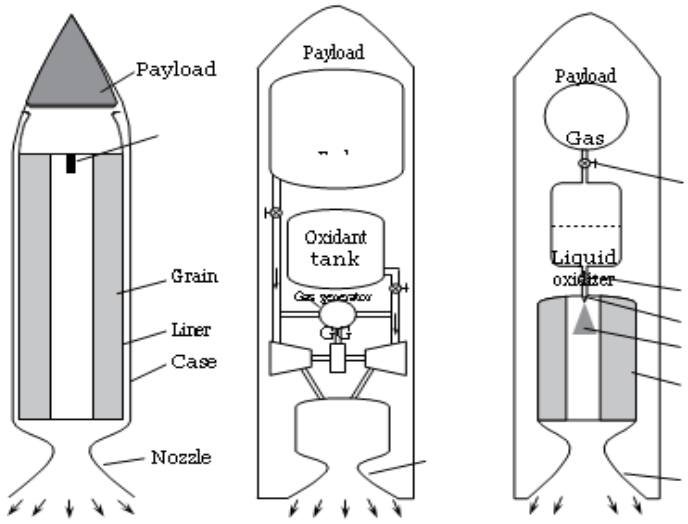


Figure 1.4: Three types of chemical rocket engines: (a) solid propellant, liquid propellant, and (c) hybrid propellant

Applications of rocket engines

In comparison to other air-breathing engines, rocket engines have very specific applications,

which can be broadly divided into four types: (1) space launch vehicle, (2) spacecraft, (3) missile, and (4) miscellaneous. Rocket engines have been used for several other civilian applications depending on the imagination of people. Rocket engines that are used routinely for collecting weather predictions are commonly known as “sounding rockets.” Rocket engines are also used for research in airplanes, in rocket-assist take off, and for providing lifelines to ships under distress. Besides this, they are considered for use in developing propulsion belts. Rockets can be used in the future for providing relief materials to inaccessible places during natural calamities.